

AI vs ML vs Deep Learning

Understanding how Artificial Intelligence, Machine Learning and Deep Learning relate

Explanation

Think of **Artificial Intelligence (AI)** as the big goal: getting computers to do things that normally need human intelligence, such as understanding language, recognising faces or making decisions. **Machine Learning (ML)** is one way to reach that goal: instead of writing every rule by hand, you show the computer lots of examples and let it work out the rules itself. **Deep Learning (DL)** is a particular kind of machine learning that uses large **neural networks** with many layers, loosely inspired by how neurons in the brain connect.

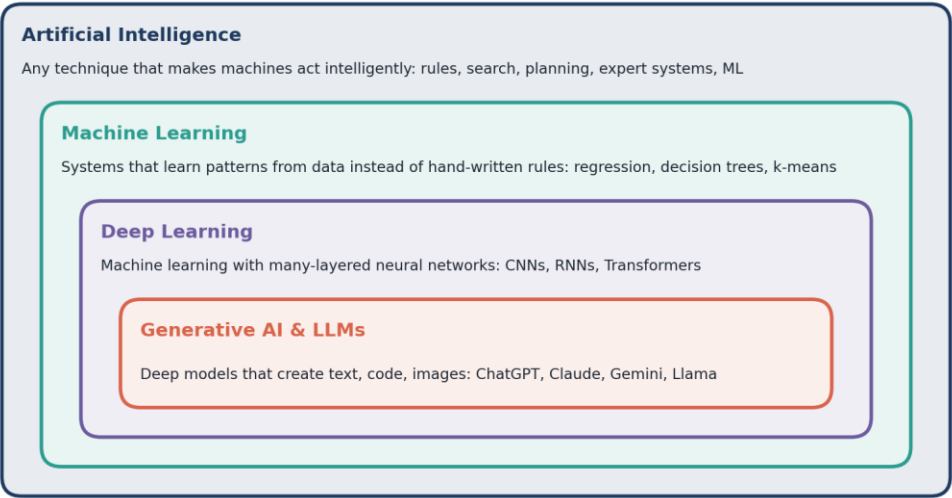
Key terms

Term	What it means
1. Artificial Intelligence (AI)	The broad field of building systems that perform tasks requiring human-like intelligence, whether through rules, search or learning.
2. Machine Learning (ML)	A subset of AI in which systems learn patterns from data rather than following explicitly programmed rules.
3. Deep Learning (DL)	A subset of ML that uses neural networks with many layers to learn representations directly from raw data.
4. Neural network	A model made of layers of connected artificial neurons, each computing a weighted sum followed by a non-linear function.
5. Training data	The examples a model learns from. The quality and quantity of this data largely decide how well the model performs.
6. Model	The result of training: a learned set of parameters that takes new input and produces a prediction or output.
7. Generative AI	Deep learning models that create new content such as text, images, code or audio, rather than only classifying or predicting.

Architecture diagram

Figure 1. How AI, ML, Deep Learning and Generative AI nest inside each other

Each layer is a subset of the one around it



Comparison table

Feature	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Definition	Broad field of making machines perform tasks requiring intelligence	A subset of AI where machines learn patterns from data	A subset of ML using multi-layer neural networks
Main idea	Make machines act intelligently	Make machines learn from data	Make machines learn complex patterns automatically
How it works	Rules, logic, search, learning, or a combination	Algorithms learn from examples and data	Neural networks learn representations from large amounts of data
Human involvement	Can require manually defined rules	Humans often select features and prepare data	Requires less manual feature engineering
Computing power	Low to high, depending on approach	Moderate	Usually high
Examples	Chatbots, game AI, expert systems, recommendation systems	Spam detection, fraud detection, credit scoring	Face recognition, speech recognition, autonomous driving
Typical algorithms	Rule-based systems, search, planning, ML	Decision trees, linear regression, random forests, gradient boosting	CNNs, RNNs, transformers, deep neural networks
Relationship	Biggest category	Subset of AI	Subset of ML

Real-world examples

Example	AI	Machine Learning (ML)	Deep Learning (DL)
1. Face Unlock on a phone	Phone identifies whether the person should be allowed to unlock	Learns patterns from examples of faces	Uses deep neural networks to recognise complex facial features
2. YouTube/Netflix recommendations	System recommends content you may like	Learns from your watching history, likes and searches	Deep models can learn complex relationships between users, videos and viewing behaviour
3. Self-driving cars	Car makes decisions such as when to stop, turn or change lanes	Learns patterns from driving data	Deep learning identifies pedestrians, vehicles, traffic signs and road conditions from cameras/sensors
4. Voice assistants	Assistant understands requests and performs tasks	Learns patterns in speech and language	Deep neural networks convert speech to text and understand language
5. Email spam detection	System decides whether an email is spam	Learns from examples of spam and legitimate emails	Deep learning can analyse complex patterns in email text and behaviour
6. Online shopping	Website provides personalised shopping experience	Learns what products you are likely to buy	Deep learning can model complex relationships between customers and products
7. Medical diagnosis	Computer assists doctors in identifying diseases	Learns patterns from previous patient data	Deep learning can analyse X-rays, CT scans and other medical images

Summary: How AI, ML and Deep Learning are connected

Artificial Intelligence (AI) is the broad field of building computer systems that can do things we normally associate with human intelligence, such as understanding language, recognising images or making decisions. There are two main ways to build these systems. The first is to write the rules by hand, as in early chess programs or expert systems. The second is to let the computer learn the rules from data. The key difference is where the system's knowledge comes from: either people program it in directly, or the system works it out from examples.

Machine Learning (ML) is the branch of AI that learns from data. Instead of writing every rule, we give the computer a large set of examples, called training data, and an algorithm adjusts the model step by step until its predictions match the correct answers as closely as possible. We then test it on new examples it hasn't seen before. This checks that it has learned general patterns rather than just memorising the training data, a problem known as overfitting. Common ML methods include decision trees, linear regression and random forests. They usually rely on people to choose which details of the data the model should pay attention to, which is called **feature engineering**.

Deep Learning (DL) is a more advanced form of machine learning that uses **neural networks** with many layers, loosely inspired by how neurons in the brain connect. Each layer picks up patterns from the one before it. In an image, for example, the early layers spot simple edges and shapes, and the deeper layers combine them to recognise faces or objects. The network finds the important features on its own, so people don't have to pick them by hand. The trade-off is that deep learning needs very large amounts of data and powerful hardware such as GPUs. Different designs suit different tasks: CNNs for images, RNNs for sequences such as speech, and Transformers, which power today's chatbots and generative AI tools.